Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



## Team Details

a. Team name: **Xcore**
b. Team leader name: **Thanveer Ahamed**
c. Problem Statement: Intelligent Risk, Fraud & Financial Experiences (BFSI). Lenders and insurers judge property risk from slow site visits or self-reported photos nobody can verify. We turn a 20-second phone video into a fraud-checked, explainable, underwriting-ready property risk file.

### Property Inspection AI

Phone video → underwriting-ready property risk file

# Brief about the idea

## From a 20-second phone video to an underwriting-ready property risk file

**Property Inspection AI** is a Gen AI inspection service for mortgage lenders, insurers and inspectors. Upload a phone walkthrough video; in minutes it returns defects, condition grade A–E, perils, repair cost, valuation impact, premium loading, an underwriting decision with reasons, fraud flags **and a 3D model that proves what was inspected.**

### THE PROBLEM: property risk is judged from photos nobody can verify

| Slow site visits | Unverifiable photos | No proof of coverage | Subjective grading | Overvaluation |
|---|---|---|---|---|
| Days of waiting; remote cases go unchecked | Reused, edited, off-screen or wrong property | Which walls did the inspection see? | Inspectors disagree; reasons unrecorded | RBI has flagged inflated collateral values |

### WHO IT SERVES: one inspection record, five BFSI decision-makers

| Lenders, HFCs, NBFCs | Insurers: underwriting | Insurers: claims | Inspectors & valuers | Owners & buyers |
|---|---|---|---|---|
| Collateral grade, valuation impact, REFER flag | Perils, risk level, premium loading | Repair-cost range, fraud-checked evidence | AI-drafted report, re-film list | Condition report, 3D model, floor plan |

**$308.6B**
yearly US insurance fraud, all lines (Coalition Against Insurance Fraud, 2022)

**~$45B**
of that in property & casualty insurance (same study)

**$4.5–12.5B**
global home-inspection services market, 2025 (industry reports)




**Minutes, not days · Fraud-resistant · Explainable**

*Concept illustration of the product*

# Opportunities

## How it is different, how it solves the problem, and our USP

### HOW DIFFERENT IS IT FROM EXISTING IDEAS?

| | Site visit / photo-only AI tools | Property Inspection AI |
|---|---|---|
| **Turnaround** | Days (visit) / minutes (tools) | **Minutes; ~10 min per 20 s video** |
| **Proof of what was inspected** | Inspector's word / none | **10 cm coverage map + 3D-pinned findings** |
| **Evidence fraud checks** | None / limited | **Keyframe dHash reuse, Gemini screen / AI check** |
| **How the decision is made** | Subjective / black-box score | **Transparent YAML rules; Gemini explains** |
| **False-positive control** | Human / single model | **Find (LocateAnything) → verify (Gemma 4) → rules** |
| **BFSI outputs & integration** | Report or damage list, PDF | **Grade, perils, cost, valuation, premium; JSON via REST API** |

### HOW WILL IT SOLVE THE PROBLEM?

| | |
|---|---|
| **Slow site visits** → | full risk file in minutes; low-risk homes clear without a visit |
| **Unverifiable photos** → | every walkthrough fraud-checked; any HIGH flag → REFER |
| **No proof of coverage** → | < 60% AI-inspected surface → REFER with a re-film list |
| **Subjective grading** → | score = 100 − Σ(points × severity × state weight) |
| **Overvaluation** → | valuation-impact %, repair retention, REFER before disbursal |

## USP of the proposed solution

**Proof of inspection.** A 10 cm coverage map and 3D-pinned findings show exactly what the AI saw, and what to film again.

**Find → verify → decide.** Two models cut false positives; deterministic rules make every decision auditable.

**Fraud-resistant evidence.** Recycled videos, screen recordings and AI footage are flagged before a decision.

**One record, five teams.** Lending, underwriting, claims, inspectors and owners share one JSON record behind a REST API.

**Under $1 per video.** ≈ $0.50 of L4 time + ≈ $0.35 of Gemini per walkthrough; the GPU Job runs only per video.

# List of features offered by the solution

## One upload gives a complete, explainable risk file

1. **Defects localised & verified**
   18 defect types on 12 components
2. **Condition grade A–E**
   0–100 score, transparent formula
3. **Risk level & insurance perils**
   Water, structural, moisture...
4. **Indicative repair-cost range**
   Per finding and per property
5. **Valuation & collateral impact**
   Impact % and repair retention
6. **Premium-loading suggestion**
   For quote and renewal
7. **Underwriting decision + reasons**
   PROCEED / CONDITIONS / REFER
8. **Evidence-integrity flags**
   Recycled video, screens, AI footage
9. **Video → 3D, floor plan, coverage**
   GLB models, room areas, maps
10. **Gemini analysis & Q&A**
    Grounded answers citing issue IDs

### ✓ Detection scope

Walls, ceilings, floor & wall tiles, grout, wood/vinyl, countertops, doors, locks, windows, trim, kitchen exhausts in 8 room types. Look-alikes (grout, grain, shadows) rejected; locks, windows, fans always manual; cracks get a structural note, water/mold a moisture check.

### ✓ Fraud & evidence integrity

Keyframe dHash reuse across every property, Gemini screen / render / AI check on every keyframe, blurred frames dropped, video coverage < 60% → REFER. Any HIGH flag → human review.

### ✓ 3D digital twin & coverage map

Depth Anything 3 + OpenCV 5 SLAM + Open3D TSDF → clean & SLAM GLB, floor plan (PNG + SVG), navigation map. Every 10 cm of surface checked; greedy AI keyframes; findings pinned to a wall and video time.

### ✓ Gemini co-pilot & product

Property analysis, grounded Q&A, scene and object naming (explains, never changes results). Portfolio dashboard, live 3D-scan view, Lite / Advanced 3D, 3D viewer, HTML / Markdown / JSON reports, REST API at /docs.

# Process flow diagram or Use-case diagram

## Capture, inspect, decide, act: minutes instead of days

**ACTORS**

- **Inspector / owner** captures
- **Underwriter / claims** reviews & decides
- **Core systems** LOS, policy, claims

**1 Capture**
A phone walkthrough video. No special hardware.

→

**2 AI inspection**
Gemini reads the room; LocateAnything finds defects; Gemma 4 verifies each crop.

→

**3 Risk decision**
Rules set grade, perils, cost, valuation, premium, decision; Gemini explains.

→

**4 Act**
Report, Gemini Q&A and a JSON record via the REST API to core systems.

**DECISION OUTCOMES (back to underwriter and core systems)**

- PROCEED
- PROCEED W/ CONDITIONS: note defects
- PROCEED W/ CONDITIONS: repairs / retention
- REFER: professional survey
- REFER: evidence integrity
- REFER: additional evidence

**KEYFRAME PIPELINE**

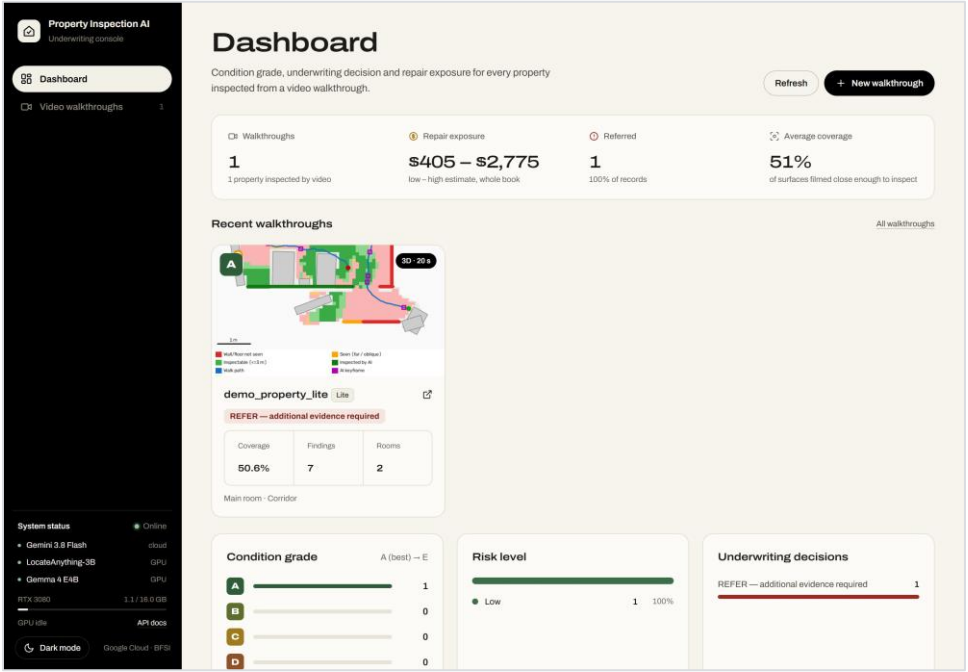
1 Quality + integrity (dHash) → 2 Gemini: room, components, authenticity → 3 Prompt router: 18 defects, 12 components → 4 LocateAnything-3B boxes → NMS → 5 Full-res crops → Gemma 4 verify → 6 Rule engine → status → risk engine

**VIDEO PIPELINE**

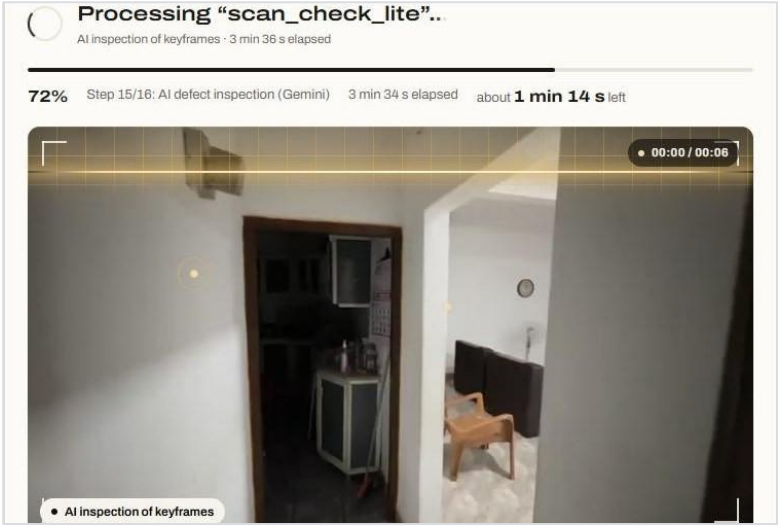
1 Depth Anything 3 depth per frame → 2 OpenCV 5 SLAM camera poses → 3 Open3D TSDF surface fusion → 4 Walls, floor, doors; Gemini names objects → 5 10 cm coverage + AI keyframes → 6 Keyframes → keyframe pipeline; pins in 3D

# Wireframes/Mock diagrams of the proposed solution (optional)

## Actual product screens: an inspection app built for underwriters




*Dashboard: walkthroughs, repair exposure, referrals, coverage and condition grades across the book*

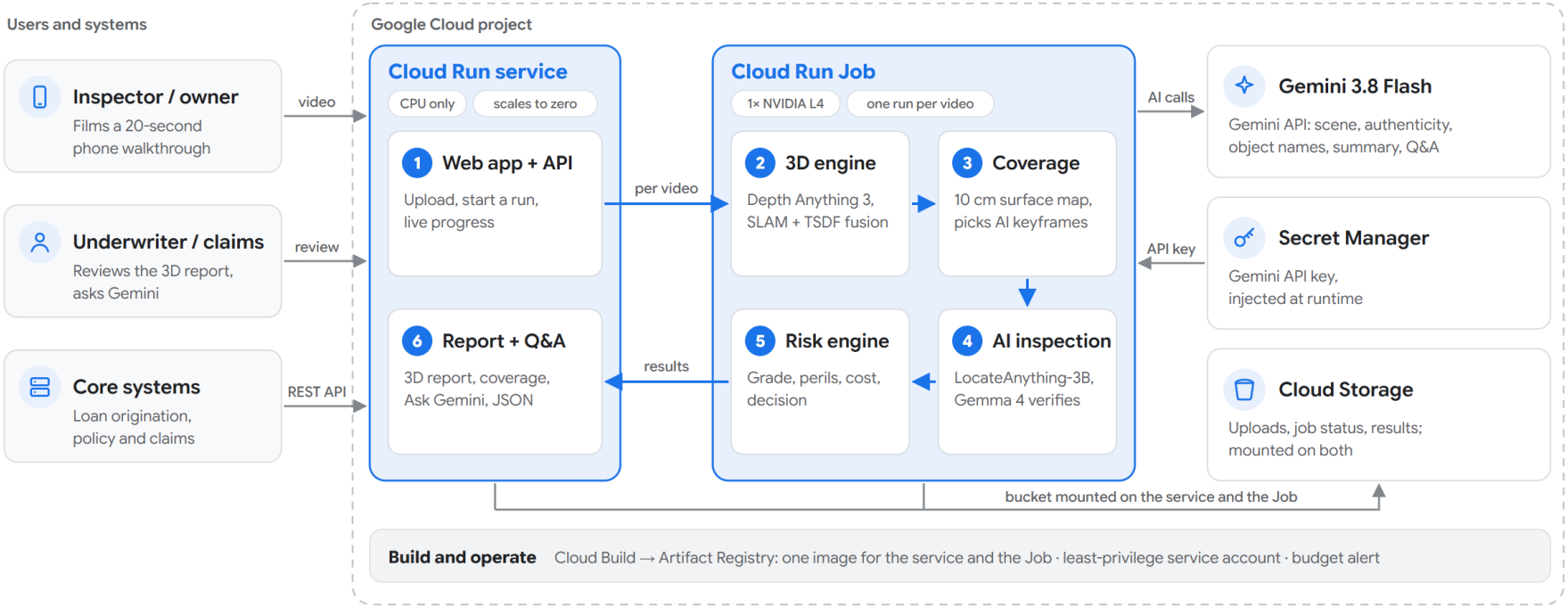



*Video walkthrough: live 3D-scan progress while the video is processed*

- Dashboard → Video walkthroughs → 3D report → Ask Gemini
- Video upload with a live 3D-scan view, Lite or Advanced
- Interactive 3D viewer: walk inside, see coverage gaps
- Reports in HTML, Markdown, JSON; REST API at /docs

# Architecture diagram of the proposed solution

## Cloud Run web service + L4 GPU Job, powered by Gemini




*Deployed layout (v1.0.0): web UI + API on a CPU-only Cloud Run service; 3D, coverage and AI inspection on a Cloud Run Job with one NVIDIA L4 per video; Cloud Storage bucket mounted on both; Gemini 3.8 Flash via the Gemini API (key in Secret Manager). BigQuery and Pub/Sub are not used.*

# Technologies to be used in the solution

## Built on Google AI and Google Cloud, with best-in-class open models

### GOOGLE AI & CLOUD, IN THE PROTOTYPE

- ✓ **Gemini 3.8 Flash** — Gemini API: scene, authenticity, analysis, grounded Q&A, 3D object names
- ✓ **Gemma 4 (E4B)** — Open Google model that verifies every defect crop on the GPU
- ✓ **Cloud Run + L4 GPU Job** — CPU web service + one L4 Job per video; both scale to zero
- ✓ **Cloud Storage** — Bucket mounted on both: uploads, job status, results
- ✓ **Secret Manager** — Gemini API key injected at runtime; none in the image
- ✓ **Cloud Build** — Builds the ~12 GB image; weights fetched inside the build
- ✓ **Artifact Registry** — One image with baked-in models for the service and the Job
- ✓ **IAM service account + Logging** — Least-privilege runtime account; no keys in files

### OPEN MODELS & LIBRARIES

- **LocateAnything-3B** open-vocabulary defect boxes
- **llama.cpp** local GPU inference for Gemma 4
- **Depth Anything 3** monocular depth per frame
- **OpenCV 5** SLAM tracking, ICP, CV evidence
- **Open3D** TSDF fusion, meshes, GLB export
- **gsplat** Gaussian-splatting experiment
- **FastAPI + Python** 18 REST endpoints, web UI, /docs
- **YAML rule engine** client-owned risk rules

### NEXT ON THE ROADMAP

- **Document AI** — Title deeds, valuation reports, repair invoices cross-checked with images
- **Vertex AI Search** — Policy wording and lending guidelines for the assistant
- **Vertex AI + BigQuery ML** — Claim-likelihood and collateral-risk models
- **Cloud SQL / AlloyDB** — Multi-tenant case management, audit trail, SSO

**Why this stack:** Gemini does reasoning, authenticity and explanation; open models on the Cloud Run GPU Job do high-volume boxing and verification cheaply; deterministic rules keep decisions auditable. Data stays in the client's own Google Cloud project, and the JSON record and REST API plug straight into lending, underwriting and claims systems.

# Estimated implementation cost (optional)

## Priced per inspection, far below a physical site visit

### UNIT ECONOMICS (INDICATIVE, FROM THE PROTOTYPE)

| Cost item | Basis | Estimate |
|---|---|---|
| Gemini calls (Gemini API) | ≈ $0.005 per call | **≈ $0.35 per walkthrough** |
| NVIDIA L4 on a Cloud Run Job | 1× L4, 8 vCPU, 32 GiB; one run per video | **≈ $0.50 per 20-min walkthrough** |
| Video inspection, all-in | L4 Job + Gemini calls | **≈ $0.85 per walkthrough** |
| Web UI + API (Cloud Run) | 1 vCPU, 2 GiB, scales to zero | **Within the free tier** |
| Idle footprint | 12 GB image + data bucket | **≈ $1.20 per month** |
| Deployment & ops | One-command deploy, $30/mo budget alert | **No servers to manage** |

**≈ $0.35**
Gemini per walkthrough

**≈ $0.50**
L4 GPU time per video

**$0**
GPU cost when idle (Job runs per video)

### BUSINESS MODEL

**Pay-per-inspection API**
Lenders and insurers send a walkthrough video from their own apps; get JSON, a report and a decision back.

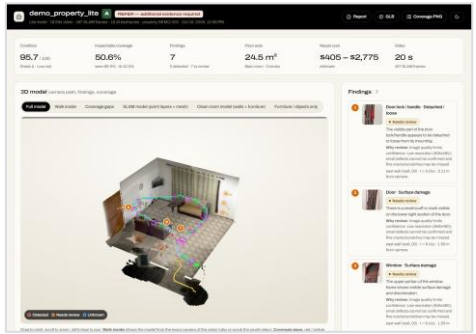
**Enterprise subscription**
Portfolio dashboard, BigQuery export, custom risk rules, SSO, private deployment in the client's GCP project.

**Inspector & valuer firms**
White-label app: AI drafts every report and coverage control, so each inspector handles more cases.

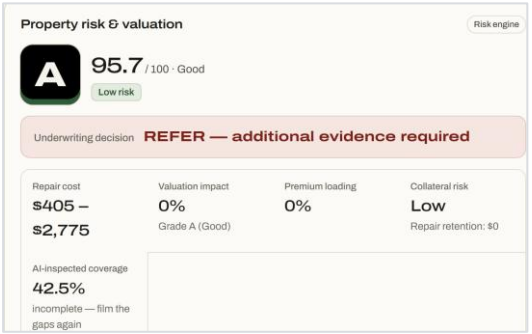
**Pricing hypothesis to validate in pilots:** a per-inspection fee well below a physical visit, with volume tiers for portfolios.

# Snapshots of the prototype

## Real screens and outputs from the working prototype v1.0.0




*3D report: findings, grade A, decision*




*Risk & valuation panel: REFER with reasons*



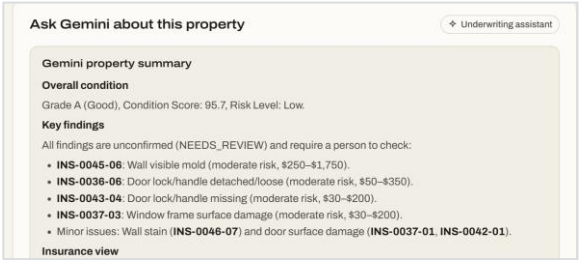

*Keyframe detection: door and window damage*



*Phone video → clean 3D model (GLB)*




*Coverage map: what to film again*




*Gemini property analysis (co-pilot)*

# Prototype Performance report/Benchmarking

## Measured on a real 20-second walkthrough, plus a fair 4-pipeline evaluation

### CASE STUDY: REAL HOME WALKTHROUGH (MEASURED)

| | | |
|---|---|---|
| **19.6 s**<br>phone video | **197**<br>SLAM frames | **15**<br>AI keyframes |
| **24.5 m²**<br>2 rooms measured | **95.7 · A**<br>condition score | **$405–2,775**<br>indicative repairs |

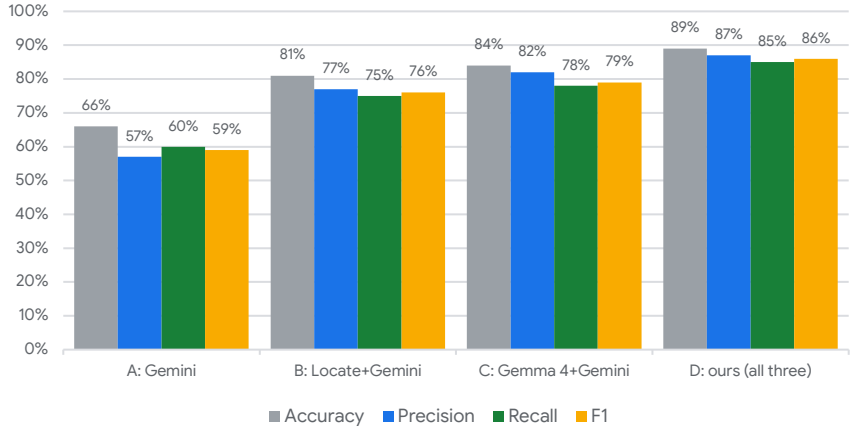
**Decision: REFER, additional evidence required**

Only 42.5% of surfaces were AI-inspected (min 60%), so the app lists the walls and ceilings to film again. 7 possible defects pinned in 3D (Needs review).

### SPEED & COST (FROM THE PROTOTYPE)

| | |
|---|---|
| **10.6 min**<br>end to end, 20 s video, laptop GPU | **11.1 / 26.9 min**<br>3D build, Lite / Advanced |
| **21.8 / 21.6 dB**<br>PSNR, Lite / Advanced | **≈ $0.35**<br>Gemini per walkthrough |

### EVALUATION: 4 PIPELINES · 10 ROOMS · 100 REGIONS




**Simulated illustration, not yet measured:** values show the report format and the expected gain; they will be replaced after running the protocol on D.

**Fair test:** same images, schema, Gemini settings and decision rule; test set frozen; 5 living rooms + 5 bedrooms incl. 2 clean controls; 40 defective / 60 clean regions.

**Targets for D:** ≥ 85% accuracy, precision, recall, F1; ≤ 1 false alarm on clean rooms; beat A on every metric.

# Additional Details/Future Development (if any)

## Roadmap: documents, policies and predictive portfolio risk

### NOW · v1.0.0

- Video walkthrough inspection
- Risk and valuation engine
- Fraud and coverage checks
- Gemini analysis and Q&A
- Cloud Run service + GPU Job

### NEXT · 3 months

- Mobile guided capture with live coverage
- Document AI: deeds, valuations, invoices
- Vertex AI Search over policy wording
- Commercial localiser (Gemini / Vertex-tuned)

### LATER · 6–12 months

- BigQuery ML claim & collateral-risk scoring
- Before/after claim comparison in 3D
- Multilingual reports (Gemini)
- AlloyDB multi-tenant case management

### WHAT WE BUILT: FROM DEPTH RESEARCH TO A DEPLOYABLE PRODUCT

**01 Depth + SLAM**
Depth Anything 3 visual SLAM, occupancy grid, A* path

**02 Hybrid SLAM**
OpenCV 5 tracking, ICP, Manhattan planes

**03 Photo-real 3D**
Gaussian Splatting experiment (gsplat)

**04 Production 3D**
Video → GLB models, floor plan (+SVG)

**05 Inspection AI**
Taxonomy, locate + verify, risk engine, fraud

**06 Google Cloud v1.0.0**
Gemini API, Cloud Run service + L4 Job

**18**
defect types, 12 components

**8**
room types

**18**
REST API endpoints

**2**
3D modes (Lite / Advanced)

**1**
command to deploy on GCP

# Additional Details/Future Development (contd.)

## Responsible AI and the go-to-market plan

### RESPONSIBLE AI: HUMANS DECIDE, AI SHOWS EVIDENCE

**1 Human in the loop**

Decision support only; a qualified surveyor or underwriter makes the final call.

**2 Explainable**

Deterministic YAML rules; every decision lists its reasons and risk points.

**3 Auditable**

Raw outputs archived; each inspection a stored JSON record; versioned config.

**4 Safe wording**

No structural-safety or root-cause claims; locks, fans, windows stay manual.

**5 Secure**

Secret Manager for the Gemini key; data stays in the client's project.

**6 Licence-aware**

LocateAnything is non-commercial; Gemini localisation is the drop-in swap.

### GO-TO-MARKET: HOUSING FINANCE & HOME INSURANCE, APAC

**PILOT · 0–3 months**

1 housing-finance lender + 1 home insurer; AI alongside existing inspections on ~500 cases.

**INTEGRATE · 3–6 months**

Pub/Sub into loan-origination and claims; BigQuery dashboards; custom rules per client.

**SCALE · 6–18 months**

Google Cloud Marketplace, valuer & inspector networks, commercial property and claims.

**Pilot KPIs:** turnaround · cost per inspection · % auto-cleared vs referred · fraud flags confirmed · re-inspection rate

***"Every lender and insurer already asks for photos. We replace them with a short phone video that is trustworthy, explainable and ready for a decision."***

Provide links to your:

- GitHub Public Repository: github.com/XCore-Lab/property-inspection-ai
- Demo Video Link (3 Minutes): [paste link here]
- Final Product Link: property-inspection-ai-77055350047.us-central1.run.app



*Property Inspection AI · Prototype v1.0.0 · one-command Cloud Run deploy*

Google Cloud H2S

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# Thank you

